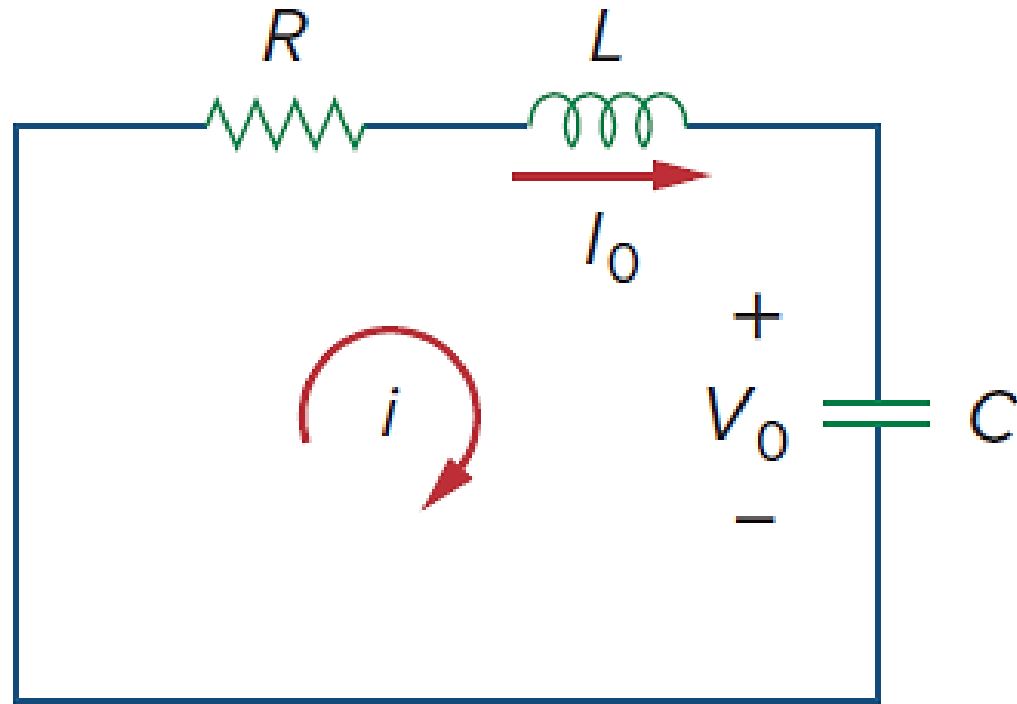


Problem 2: For the given circuit, $R = 40 \, \Omega$, $L = 4 \, \text{H}$, and $C = 1/4 \, \text{F}$. Calculate the characteristic roots of the circuit. Is the natural response overdamped, underdamped, or critically damped?



$$R = 40 \, \Omega, L = 4 \, \text{H}, C = 1/4 \, \text{F} \Rightarrow \alpha = \frac{R}{2L} = 5; \omega_0 = \frac{1}{\sqrt{LC}} = 1$$

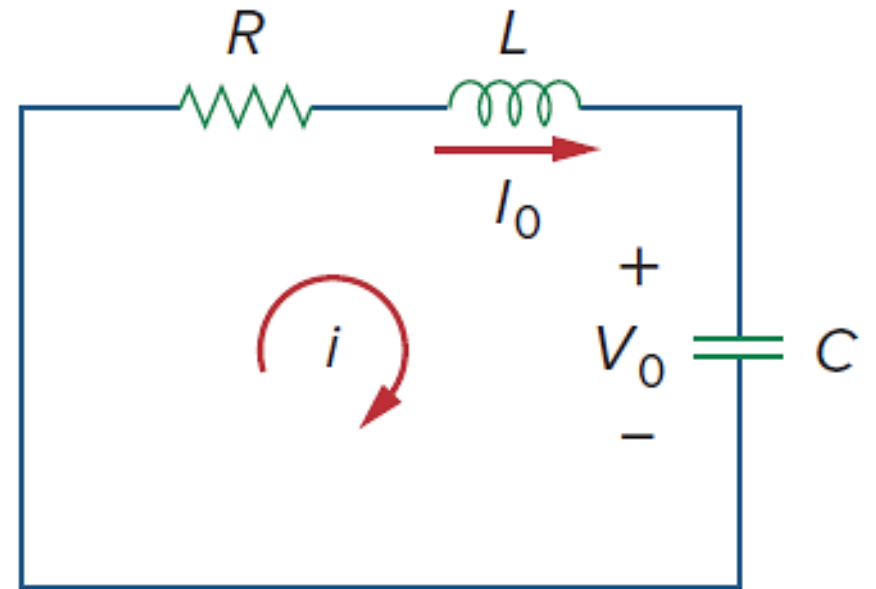
$$\text{The roots are, } s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -5 + \sqrt{5^2 - 1}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -5 - \sqrt{5^2 - 1}$$

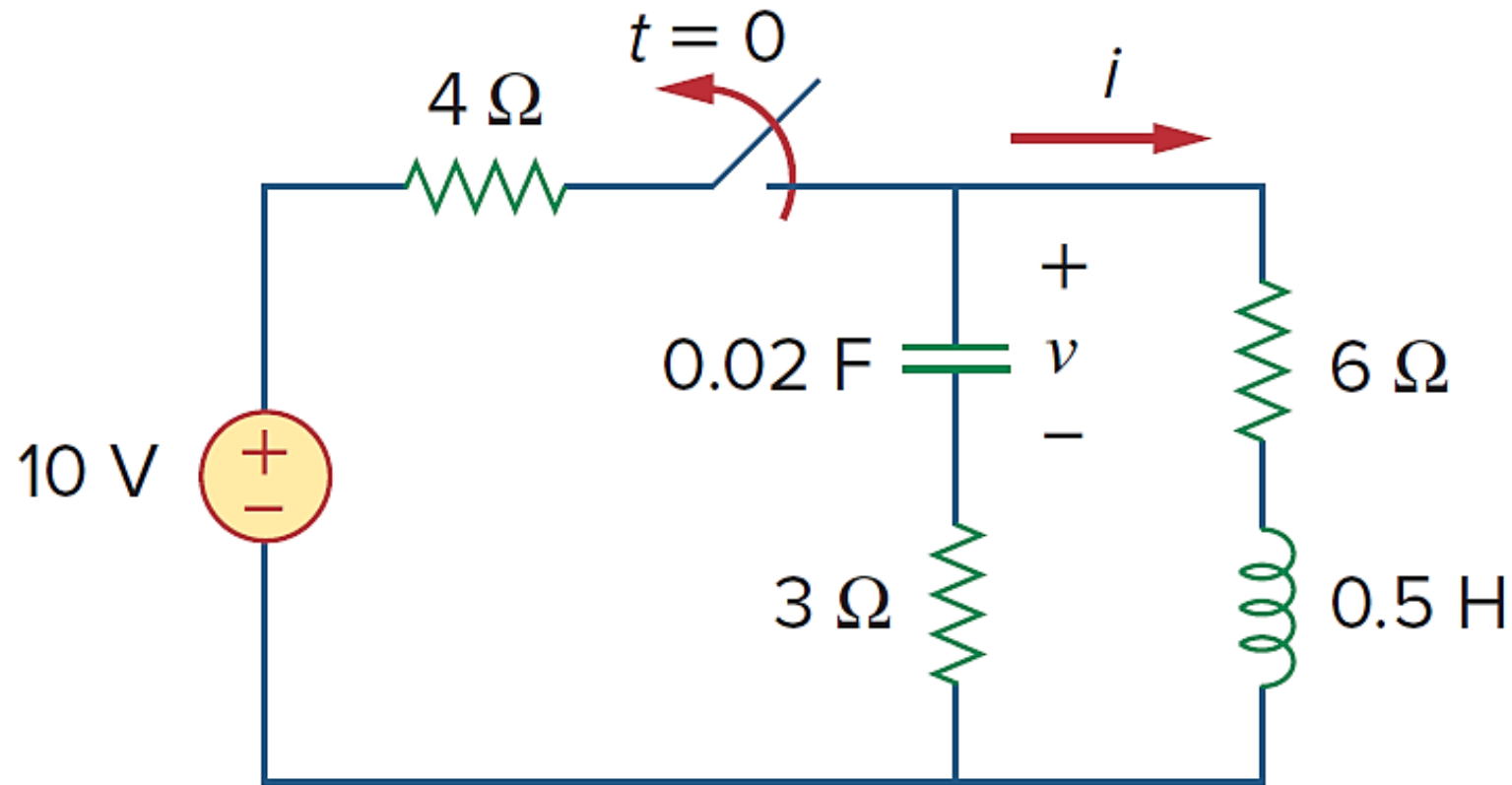
$$s_1 = -0.101; s_2 = -9.899$$

Since $\alpha > \omega_0$, we conclude that the response is **overdamped**.

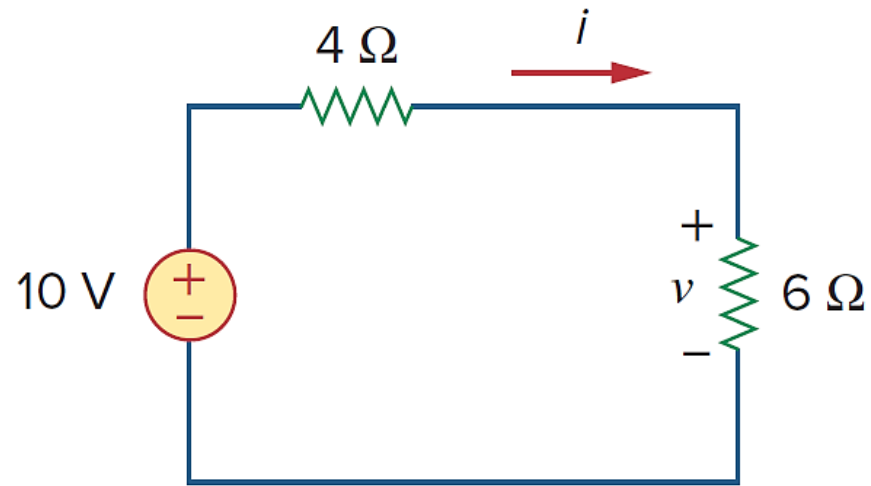
This is also evident from the fact that the **roots are real and negative**.



Problem 3: Find $i(t)$ for $t > 0$. Assume that the circuit has reached steady state before the switch is opened.

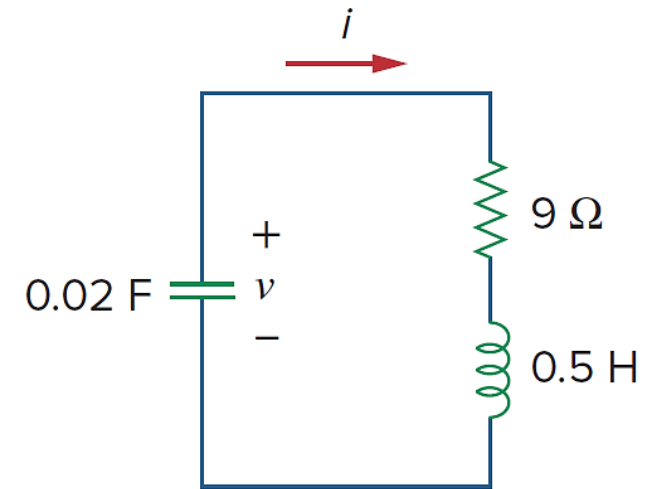


Solution:



(a) for $t < 0$

$$i(0) = \frac{10}{4 + 6} = 1\text{A}; \quad v(0) = 6i(0) = 6\text{V}$$



(b) for $t > 0$.

$$\alpha = \frac{R}{2L} = 9; \quad \omega_0 = \frac{1}{\sqrt{LC}} = 10$$

$$s_1 = -9 + \sqrt{9^2 - 100}$$

$$s_2 = -9 - \sqrt{9^2 - 100}$$

Under-damped response

$$s_{1,2} = -9 \pm j4.359$$

Hence, $i(t) = e^{-9t}[A_1(\cos 4.359t) + A_2(\sin 4.359t)]$

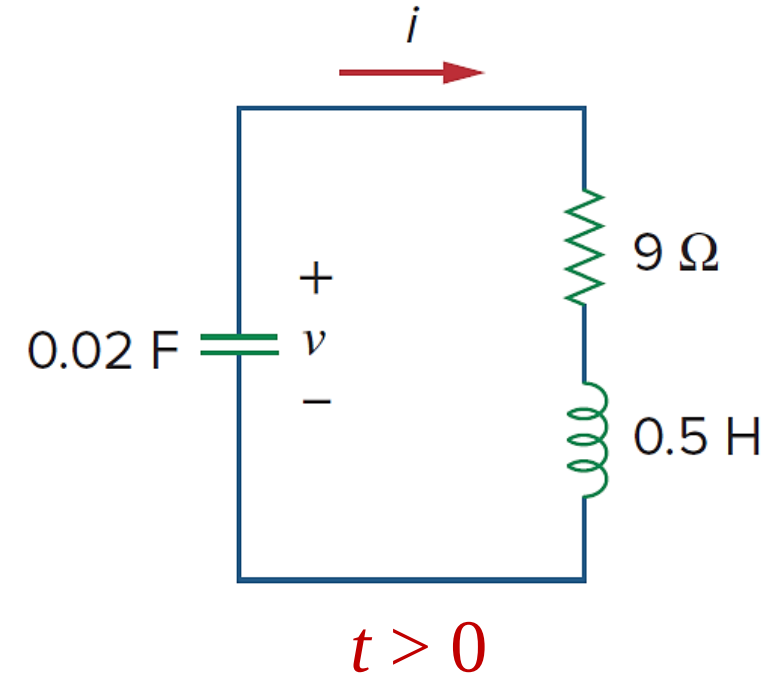
A_1 and A_2 are found using the initial conditions.

At $t = 0$, $i(0) = 1 = A_1$

$$\left. \frac{di}{dt} \right|_{t=0} = -\frac{1}{L}[Ri(0) - v(0)] = -6 \text{ A/s}$$

Taking the derivative of $i(t)$

$$\begin{aligned} \frac{di}{dt} = & -9e^{-9t}[A_1(\cos 4.359t) + A_2(\sin 4.359t)] \\ & + e^{-9t}(4.359)[-A_1(\sin 4.359t) + A_2(\cos 4.359t)] \end{aligned}$$

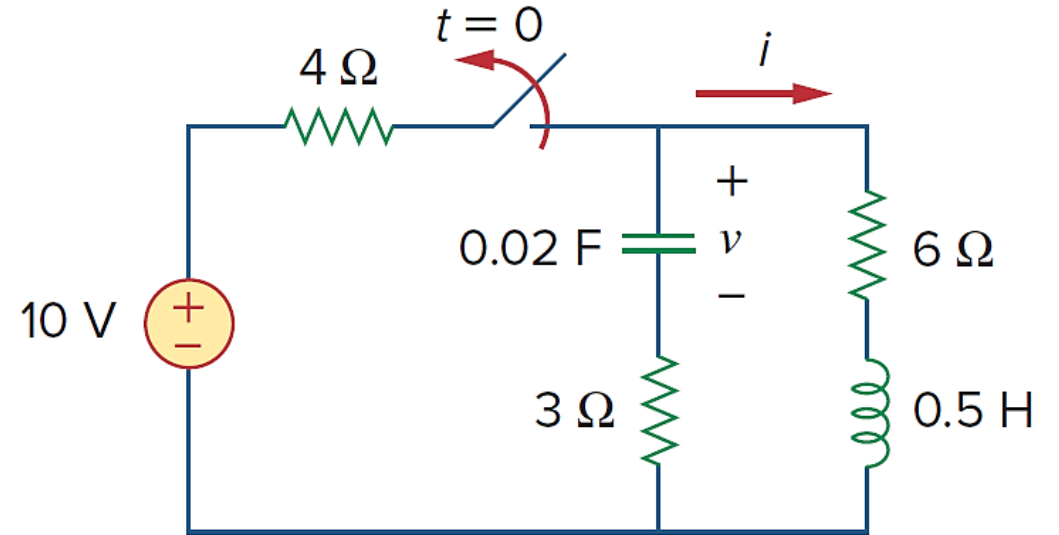


$$-6 = -9(A_1 + 0) + 4.359(-0 + A_2)$$

substituting $A_1 = 1$,

$$-6 = -9 + 4.359A_2$$

$$\Rightarrow A_2 = 0.6882$$



Substituting the values of A_1 and A_2 yields the complete solution as,

$$i(t) = e^{-9t}[(\cos 4.359t) + 0.6882(\sin 4.359t)] A \quad \text{for } t > 0$$